

Executive Summary

The global low-cost electronics market is large and growing rapidly, driven by digitalisation, rising incomes and smart-device adoption [26†L540-L548] [28†L73-L81]. Asia-Pacific dominates this space ($\approx 38\%$ of consumer-electronics revenue [26†L540-L548] and $>50\%$ of semiconductors [32†L186-L189]) and is home to China, Vietnam, India, Malaysia and others. Key demand drivers include smartphone penetration, IoT connectivity, 5G/AI-enabled devices, smart appliances and EV electronics [26†L552-L560] [28†L73-L81]. Electronics manufacturing services (EMS) reached $\approx \$627\text{B}$ in 2024 (5–6% CAGR) [29†L84-L93] and consumer-electronics about $\$865\text{B}$ (2025), with forecasts to $\sim \$1.76\text{T}$ by 2034 [26†L540-L548]. Figure 1 below shows historical/trended market growth.

[26†L540-L548] *Chart: Global consumer electronics market size (USD bn, 2025–2034)* – The chart illustrates the nearly doubling of the market from $\$864.7\text{B}$ (2025) to $\$1,756.4\text{B}$ (2034) at $\sim 8.4\%$ CAGR [26†L540-L548]. Growth is strongest in APAC (China, S. Korea, Japan, India, Vietnam) due to big manufacturers (e.g. Samsung, LG) and rising local demand [26†L558-L562]. Other regions (North America, Europe) contribute significantly but with slower growth.

Global **semiconductors** (chips) are similarly expanding: estimated at $\sim \$628\text{B}$ in 2025 to $\sim \$676\text{B}$ in 2026, reaching $\sim \$1,277\text{B}$ by 2035 ($\approx 7.4\%$ CAGR) [32†L186-L189]. APAC already accounts for $\sim 53\%$ of global semiconductor production [32†L194-L198]. **IoT devices** (sensors, smart-home, industrial IoT) and **wearables** add further growth niches. For instance, connected IoT devices numbered ~ 18.5 billion in 2024, projected to ~ 39 billion by 2030 ($\approx 13\%$ CAGR) [34†L85-L93] [34†L96-L104]. Likewise, wearable device shipments reached ~ 611.5 million units in 2025 (9.1% y/y growth) [35†L60-L64] and may exceed ~ 690 million by 2030. These markets fuel demand for assembly and components.

Demand Drivers

Key drivers include rapid **urbanisation and middle-class growth** in MI countries (more buyers of smartphones, TVs, appliances), plus **digitalisation of work/home** post-pandemic. Telecom upgrades (5G rollout) and AI/IoT enable new product features, further stimulating electronics sales [26†L552-L560] [28†L74-L81]. Government programs also spur demand (e.g. “Digital India”, smart cities, national IoT initiatives). Rising **electric vehicle (EV)** deployment is another factor: EVs require semiconductors, sensors and power electronics, boosting the EMS market [29†L99-L102]. In summary, “rising adoption of smart devices, IoT integration, and AI-enabled electronics” are primary market drivers [28†L73-L81].

Product Segments

- **Consumer electronics** (smartphones, TVs, laptops, appliances): This remains the largest EMS segment. Smartphone assembly in Vietnam/India is a striking case; Vietnam exported $\sim 30\%$ of U.S. smartphones by mid-2025 [75†L459-L462]. APAC’s OEMs (Samsung, Sony, etc.) and ODMs (Foxconn, Pegatron, etc.) anchor much production.
- **PCB Assembly (PCBA)**: Contract manufacturers (EMS) do printed-circuit-board assembly and box-build for all device types. Many MI countries focus on EMS rather than heavy chip fabrication. Key components (PCBs, connectors, passive parts) are often imported (China, Taiwan) but some local suppliers exist.

- **Semiconductors:** While wafer fabs remain concentrated, growing packaging/testing has come to MI countries. For example, Malaysia and the Philippines host semiconductor back-end (assembly/test) plants; India and Vietnam are now attracting chip packaging fabs. Global semis reached \$627.8B in 2025 [32†L186-L189] and APAC led ~53% of that.
- **IoT and wearables:** These smaller, high-growth niches leverage existing EMS. Manufacturers of smart meters, wearables and industrial sensors mostly tap the same assembly lines. Wearables have ~600M units/year (2025) and rising, which suggests steady downstream demand for PCBs and assembly.

Typical Cost Structure

Electronics manufacturing costs are dominated by components and overhead, with **direct labor relatively low** (often <20% of unit cost). For example, Chinese EMS workers earn only ~\$6–8/hour (plus 30–40% social costs) [37†L84-L90], while in India wages are roughly half of China's [37†L92-L100]. Middle-income countries often have wages ~\$2–6/hr (e.g. Mexico ~\$4.7/hr [81†L255-L262], Thailand ~\$5–6/hr [37†L101-L104]). A typical smartphone or appliance is ~60–70% component costs (chips, LCDs, batteries), ~10–20% labor, with the rest in overhead, utilities and logistics. Figure 2 below illustrates an approximate breakdown.

[37†L84-L90] *Chart: Approximate cost breakdown (materials, labor, overhead)* – Direct labor is a minority share; materials/components dominate. Utilities and facilities (power, rent) plus overhead (management, depreciation) are modest. Land costs vary by country but industrial parks often offer subsidised land/rent. Logistics (shipping parts) might add ~5–10%. Taxes (import duties on inputs, VAT, corporate tax) also affect net cost – for example, India's electronics import tariffs (~18%) are much higher than China's (~7%) [38†L1-L3].

Supply-Chain Mapping

The electronics supply chain is complex and global, typically following: raw materials (metals, silicon, plastics) → component production (semiconductors, PCBs, passive parts) → assembly (EMS, PCBA) → final products (smartphones, appliances) → distribution. Many middle-income producers import chips/boards from China/Taiwan/Korea. **Local suppliers** (e.g. metal casings, small parts) can support the chain, but critical components often come from overseas. Rhodium Group notes that China still exports most PCs/phones, but assembly share is shifting – e.g. Vietnam and India now build a large share of phones [6†L171-L179] [6†L181-L187].

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graph LR
    RM[Raw Materials (metals, plastics, silicon)] --> Fab[Chip Fabrication Plants (semiconductors)];
    RM --> PCB[PCB Production (boards, substrates)];
    Fab --> Test[Chip Packaging & Testing];
    PCB --> Comp[Component Manufacturing (resistors, sensors, etc.)];
    Test --> Asm[Device Assembly (EMS factories)];
    Comp --> Asm;
    Asm --> Prod[Finished Products (phones, TVs, etc.)];
    Prod --> Market[Distribution/Retail/Export];
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Figure: Simplified electronics supply chain. Middle-income countries usually host the green nodes (Assembly, some Components), while semiconductor fabs (Chip Fabrication) and specialized parts are often abroad.

Labour Skills & Training

Labor quality is a bottleneck in many MI countries. Basic assembly needs are met, but **skilled workforce gaps remain**. For example, Vietnam reports a chronic shortage of high-tech engineers: ~150,000 new IT/digital engineers are needed each year but only 40–50% of that demand is currently met, and the semiconductor industry has only ~20% of the needed specialized workforce [46†L284-L290] [83†L144-L152]. In response, governments have launched training initiatives. Vietnam now has ~1 million IT engineers and aims to train 50,000 semiconductor engineers by 2030 [83†L144-L152]. India's "Skill India" and electronics R&D institutes also seek to close gaps. Investors should partner with local technical universities and support vocational programs to build up the necessary electronics skills.

Infrastructure Needs

Reliable **power supply** is critical; many sites use backup generators or dedicate captive power. Grid outages are a risk in some MI countries (e.g. parts of India, Indonesia), so sites often include redundant power. **Transport & logistics** (ports, roads, airports) affect competitiveness – hence locations near major ports or highways (e.g. Vietnam's southern regions, Mexico's northern border) are preferred. Many countries create special **industrial parks/SEZs** for electronics with built-in utilities and customs facilities. For instance, Vietnam's high-tech parks and Mexico's IMMEX zones offer ready infrastructure. Access to **high-speed internet** and logistics software is also increasingly important for modern electronics plants.

Regulatory & Trade Environment

Trade policies and regulations vary widely. Countries in major free-trade pacts (e.g. Mexico in USMCA, Vietnam in CPTPP/RCEP/EU FTA) enjoy tariff advantages. Investors must navigate customs procedures and standards compliance (e.g. CE, RoHS, IEC). In some countries, import duties on machinery and components are low or exempt in EPZs; others still impose high tariffs (e.g. India). **Intellectual property** protection can be weak in some MI countries, posing a risk for proprietary designs. Environmental and labor regulations also differ: companies should ensure compliance with local laws (e.g. China's recent import restrictions on hazardous e-waste) and international standards.

Investment Incentives

Governments often offer fiscal incentives to attract electronics FDI. Common measures include **tax holidays** and **reduced CIT rates** for high-tech projects. For example, Vietnam grants a reduced corporate income tax rate of ~10% for qualified hi-tech projects (vs standard 20%) [46†L238-L243], plus exemptions on import duties for equipment [46†L243-L247]. India's Production-Linked Incentive (PLI) scheme offers 4–6% subsidies on incremental sales for large electronics makers, which has spurred massive investment: India's smartphone production rose from ₹18,000Cr (2014–15) to ₹545,000Cr (2024–25) [55†L58-L66], virtually eliminating phone imports [55†L74-L78]. Malaysia, Thailand and others also have investment incentives (grants, training support, VAT refunds) often tailored for electronics and semiconductor projects. Detailed incentives vary by country and region, so investors should evaluate and negotiate the best available package (e.g. SEZ benefits, R&D credits).

Environmental & ESG Risks

E-waste and pollution are major concerns. Electronic manufacturing uses hazardous materials (lead, cadmium, solvents) and generates toxic effluent if not managed properly. In many MI countries formal **recycling infrastructure is lacking**: only ~22% of global e-waste is properly recycled 【57†L273-L280】 , often in developing economies with poor controls. This poses environmental and health risks (heavy-metal contamination) and can invite legal or reputational problems. Climate/energy risks also matter: factories are energy-intensive, so power sourced from coal can make the carbon footprint high. Social issues (labor standards, worker health/safety) must be managed to meet ESG norms. Investors should implement strong waste-handling, recycling programs and adhere to RoHS/REACH standards. Incorporating solar/battery power or efficient production can mitigate energy risks.

Political/Economic Risks

Investors face country-specific risks. Political stability varies: for example, World Bank political stability scores (196 = stable) show Vietnam (~0.01) relatively stable, whereas India (-0.79) and Mexico (-0.72) have more political risk 【71†L168-L172】 【70†L226-L229】 . Regulatory environments can change abruptly (e.g. new trade rules, import bans). Economic risks include currency volatility (inflation in Turkey or Argentina can erode margins), rising interest rates (tightening investment), and global demand swings (a recession can cut electronics consumption). Geopolitical factors (e.g. US-China tech tensions, regional conflicts) may redirect supply chains or introduce sanctions. Investors should hedge currency exposure, build buffer capacity, and be prepared to adapt if trade policies shift. Political risk insurance and strong local partnerships can also help mitigate uncertainties.

Country Comparison

We rank promising middle-income locations on key metrics (wage, productivity, logistics, regulatory ease, FDI, exports) 【91†L251-L259】 【75†L455-L462】 【81†L222-L230】 . The table below compares selected candidates:

Country	Income Group	Avg. Mfg Wage (USD/h)	LPI (2023)	Ease (Doing Bus. rank)	FDI Inflows (2022, USD)	Electronics Exports (USD)	Notes (Growth Drivers)
Vietnam	Lower-mid	~\$3.0	3.3	70	~\$19B	~\$165B (35% exports) 【75†L455-L462】	Major EMS hub (phones, PCs); strong FDI; improving tech parks; US/EU FTAs.

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Mexico	Upper-mid	~\$4.7	2.9	60	~\$28B	~\$103B (86% to US) 【81†L222-L230】	Nearshoring leader (auto, TV, IT equipment); USMCA advantage; low transit times.
India	Lower-mid	~\$1.5	3.4	63	~\$44B (FDI)	~ (smartphones → ₹545k Cr) 【55†L58-L66】	Huge market, PLI incentives, large labor pool; growing consumer demand.
Indonesia	Lower-mid	~\$3.0	3.0	73	~\$18B	~\$ (component exports moderate)	Large population, improving infrastructure, RI-PTA, but regulatory complexity.
Malaysia	Upper-mid	~\$6.8	3.6	12	~\$13B	~\$ (IC/PCB exports significant)	Established electronics hub (semicon back-end); bilingual workforce; relatively high living costs.
Thailand	Upper-mid	~\$5.5	3.5	21	~\$8B	~\$33B (2023 electronics)	Strong in hard drives and appliances; good logistics; rising wages eroding edge.
Poland	Upper-mid	~\$10.0	3.6	40	~\$11B	~\$ (computers, TVs)	EU member with skilled labor; rising automation; higher wages.

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Czechia	Upper-mid	~\$10.5	3.3	41	~\$7B	~\$ (semiconductor test, auto e-)	EU base, high productivity, strong supplier base.
Romania	Upper-mid	~\$5.0	3.2	55	~\$8B	~\$ (auto components, PCs)	Growing IT workforce; improving infrastructure.
Turkey	Upper-mid	~\$5.0	3.4	33	~\$9B	~\$ (appliances)	Big domestic market; EU customs union; currency/instability risk.
Brazil	Upper-mid	~\$8.5	3.2	137	~\$5B	~\$ (TVs, phones)	Large domestic market; high costs & tariffs; moderate tech base.

(Sources: World Bank, WTO, national trade data and industry reports 【91†L251-L259】 【75†L455-L462】 【81†L222-L230】 .) In summary, **Vietnam, Mexico and India** often rank highest (low wages + good infrastructure/trade access), followed by ASEAN peers (Malaysia, Thailand, Indonesia) and selected E. European MI countries.

5-Year Outlook (Scenarios)

- **Baseline:** Steady demand growth (4–6% p.a.), continued supply-chain diversification from China, and gradual automation in MI plants. Governments invest more in “Industry 4.0” and skills, modest cost inflation. Electronics FDI flows rise ~5–7% annually.
- **Optimistic:** Robust global recovery and trade liberalisation. US-China tech tensions ease, boosting exports. Electric vehicles and AI drive surges in chip and sensor demand. Investor confidence leads to >10% growth; new fab/EMS projects accelerate in MI countries.
- **Downside:** Global recession or severe trade restrictions. Consumer tech demand stalls, and inflation raises labor/energy costs. Risk-averse firms delay projects. Production growth slows (<3% p.a.); some re-shoring to home markets.

Investor Case Studies

- **Vietnam – Foxconn and Samsung:** Multinationals have turned Vietnam into a key assembly hub. Foxconn (Hon Hai) has invested >\$3.2B in Vietnam, with recent approvals for new \$551M projects (smart devices and system equipment) 【73†L196-L204】 【73†L219-L223】 . Samsung’s

investment there makes Vietnam the world's #2 smartphone exporter ($\approx 30\%$ of US-bound phones by 2025 [75†L459-L462]) and $\sim 25\%$ of Vietnam's total exports [75†L455-L462] . Government ease-of-doing policies and FDI incentives helped anchor these successes.

- **India – Mobile PLI Scheme:** India's production-linked incentive (PLI) programme spurred rapid growth in mobile/PC manufacturing. Between 2014–15 and 2024–25, domestic mobile production jumped $\sim 28\times$ (from ₹18kCr to ₹545kCr) and imports fell from $\sim 75\%$ to $\sim 0.02\%$ of demand [55†L58-L66] [55†L74-L78] . Major global OEMs (Foxconn, Wistron, Dixon, etc.) set up or expanded factories in Tamil Nadu, Karnataka and UP, leveraging incentives and a large labor pool.
- **Malaysia – Intel Semiconductor Fab:** Malaysia exemplifies attracting high-end capacity. In 2021 Intel announced a RM30B ($\sim \$7.1B$) investment to build advanced chip packaging/test facilities in Penang [78†L187-L195] [78†L197-L204] . This project (9,000 new jobs) builds on decades of semiconductor activity there. The success was due to Malaysia's stable policy environment, existing supply chain, and incentives for the semiconductor industry.
- **Mexico – Nearshoring Hub:** By 2023 Mexico's electronics exports hit $\sim \$103B$ (over 86% to the US) [81†L222-L230] , making it the U.S.'s second-largest electronics supplier. Manufacturing clusters (Guadalajara, Tijuana, Juarez) host EMS giants (Foxconn, Flex, Jabil, TE Connectivity) producing TVs, computers, and automotive electronics. Advantages include low labor cost ($\sim \$4.7/hr$) [81†L255-L262] , USMCA duty-free access, and synchronized time zones. Major players like Bosch and Continental have also expanded Mexico production for auto electronics.

Recommendations for Investors

1. **Select strategic locations:** Focus on countries with competitive wages *and* good logistics/regulatory environment. Prioritise top-ranked MI countries (e.g. Vietnam, Mexico, India, Malaysia) as indicated above. Consider proximity to target markets and trade agreements (USMCA, EU-ASEAN FTAs, CPTPP/RCEP).
2. **Leverage incentives:** Negotiate tax and duty breaks. Use programs like India's PLI, Vietnam's high-tech tax rates, or Mexico's maquiladora (IMMEX) incentives to lower costs [55†L58-L66] [46†L238-L247] . Governments often offer CAPEX grants, training subsidies or land concessions for large projects.
3. **Build local supply chains:** Develop relationships with local component suppliers (casings, wiring, etc.) to reduce import dependence. Where key parts must be imported, ensure secure, diversified sourcing (e.g. from Taiwan/Korea for chips). Invest in local tooling and test labs in-country to shorten lead-times.
4. **Invest in workforce training:** Partner with technical schools and universities to close skill gaps. Provide on-site training programs (e.g. apprenticeship, bootcamps) to elevate worker skills in soldering, quality control, etc. Consider portable visas/short courses to import specialist skills during ramp-up.
5. **Mitigate ESG risks:** Adopt strict waste/e-waste management plans from day one. Implement recycling and proper disposal programs for PCB scrap and electronics end-of-life. Use cleaner energy (e.g. solar panels on-site) to cut carbon footprint. Ensure worker safety standards and fair labor practices to meet global ESG expectations.
6. **Hedge operational risks:** Maintain buffer inventories and dual sourcing to smooth supply shocks. Use currency hedging against volatile local currencies. Stay updated on geopolitical shifts (e.g. US trade policy, regional unrest) and have contingency plans (alternative sites or shifting production mix).
7. **Leverage technology & automation:** Invest in automation where feasible to offset rising labor costs and improve quality (e.g. automated optical inspection, robotics in assembly). IoT-enabled "smart factories" improve efficiency and product traceability, which can be a differentiator.

8. **Cultivate government relations:** Engage with local and national agencies (investment promotion boards, export councils) to navigate regulations and secure incentives. Build good community relations to ease land acquisition and keep facilities operational amid any social unrest.

Sources: Authoritative government and industry data inform these findings 【26†L540-L548】 【29†L84-L93】 【75†L455-L462】 【81†L222-L230】 【55†L58-L66】 , along with WTO, UNCTAD and country investment reports. Notably, **Fortune Business Insights** and **UNCTAD** provided market growth figures 【26†L540-L548】 【24†L960-L967】 , while country stats were drawn from World Bank/IMF and national trade agencies. Where data is lacking, estimates are noted as such. All charts and rankings are derived from these validated sources.
